Boston College || Woods College of Advancing Studies || ADAN|ADEC7900 Software Tools

R Homework 1

Objective

This assignment is about learning the basics of R syntax, logic and structure. Your goal will be to write chunks of code that utilize common commands and operators you will be using extensively later in other assignments.

Instructions

1. You should submit this assignment as a single R script file that produces all the necessary output as required by the questions below. Important: your entire code should run without any errors!
2. Start with the provided template.
3. Comment thoroughly
4. Use only base R functions. Do not use functions from libraries installed.
5. Make sure there is no redundant/unused code in your files. Even if you use extra code to debug your work, it should be cleaned up before submitting.

FizzBuzz

In this question you will need to replicate a common group word game for children called "FizzBuzz." Every computer science student loves FizzBuzz.

1. Define a function `fizz(n)` that takes a whole number n and returns a character string "Fizz" if n is divisible by 3 and an empty string `""` if it is not.

Use `%%` operator to calculate a remainder from a division, e.g. `7 %% 5`

Hint: Avoid using `print()` to display a result. Instead, use return() . They look similar, but returning a variable to the R environment allows you to save it as a global variable or input it to another function for further processing.

2. Define a similar function `buzz(n)` that takes a whole number n and returns a character string `"Buzz"` if n is divisible by 5 and an empty string `""` if it is not.

3. Use the two functions above to define a function `fizzbuzz(n)` that takes a whole number n and returns one of the following results:

- a character string `"Fizz"` if n is divisible by 3
- a character string `"Buzz"` if n is divisible by 5
- a character string `"FizzBuzz"` if n is divisible by both 3 and 5
- an empty character string `""` if none of the above

There should be no conditional statements inside `fizzbuzz(n)` checking for remainder of the division (e.g. if (`n %% 5 == 0`)). Instead, you should call your previously defined functions `fizz(n)` and `buzz(n)` from inside `fizzbuzz(n)` to evaluate the supplied whole number argument. You can still use conditional statements to check for other things, such as whether the output of `fizz(n)` is equal to `"Fizz"` or not.

Hint: the whole `fizzbuzz(n)` code can be written as one simple line using paste() function to combine the outputs of `fizz()` and `buzz()` .

4. Use the function `fizzbuzz()` to define a function `fbr(n,m)` that takes two whole numbers n and m as two inputs, goes through all whole numbers from n up to m, returning a vector that for each number between n and m contains either a string `"Fizz/Buzz/FizzBuzz"` as defined by `fizzbuzz(n)` or the number itself.

Example: `fbr(10,15)` should return the following vector:

`( " Buzz" , " 11" , " Fizz" , " 13" , " 14" , " FizzBuzz" )`

4.1 (Optional, not graded) Make `fbr(n,m)` check whether supplied arguments are integer variable type (using `is.integer()` function) and print an error message (using `stop()` function) otherwise.

4.2 (Optional, not graded) Make `fbr(n,m)` check whether n is less or equal to m and print an error message (using `stop()` function) otherwise.